VIrtual Twins as tools for personalised clinicAL care

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0D calibration/Deep learning
Tutorial session

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Education and Research EAER
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Research and Innovation SERI

UK Research and Innovation

Tutorial outline

Part I

- Basic hemodynamic analysis of P,Q signals
- Estimation of compliance using the Pulse Pressure Method
- 2-element, 3-element Windkessel models

Part II

- An example of deep learning techniques to assess age from PPG signals

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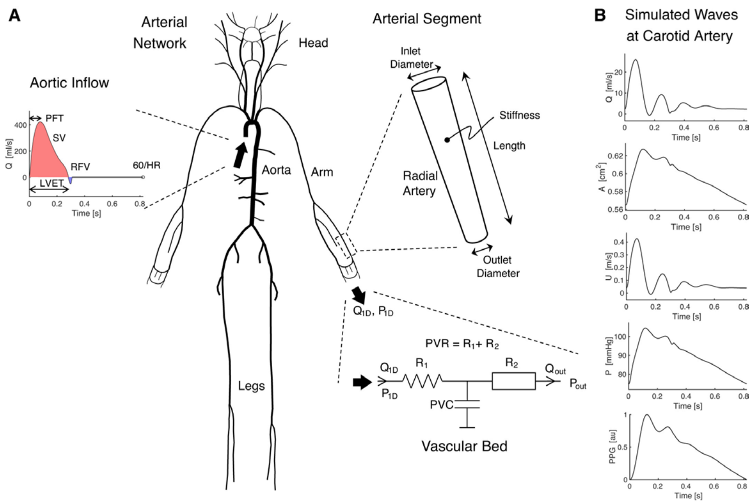
Important!

- Please download the Pulse Wave Data Base (PWDB) .mat file by P. Charlton and place it in the 'data' folder!

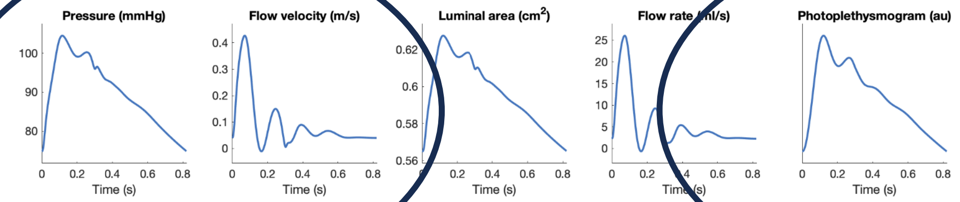
A publicly available synthetic database of pulse waves based on a 1D model



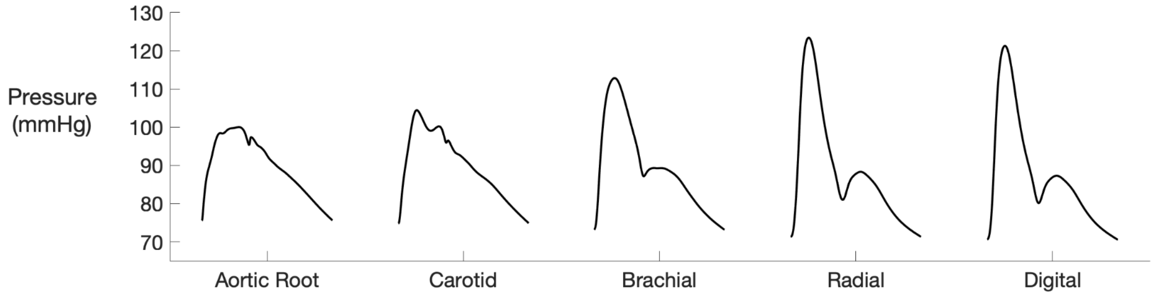
Charlton … Alastruey, *AJP* 2019

Part I: P,Q signals

Part II: PPG



Different types of pulse waves: The database contains pressure, flow velocity, luminal area, flow rate and photoplethysmogram (PPG) pulse waves.



Pulse waves at different anatomical sites The database contains pulse waves simulated at a range of anatomical sites, including 13 common measurement sites.

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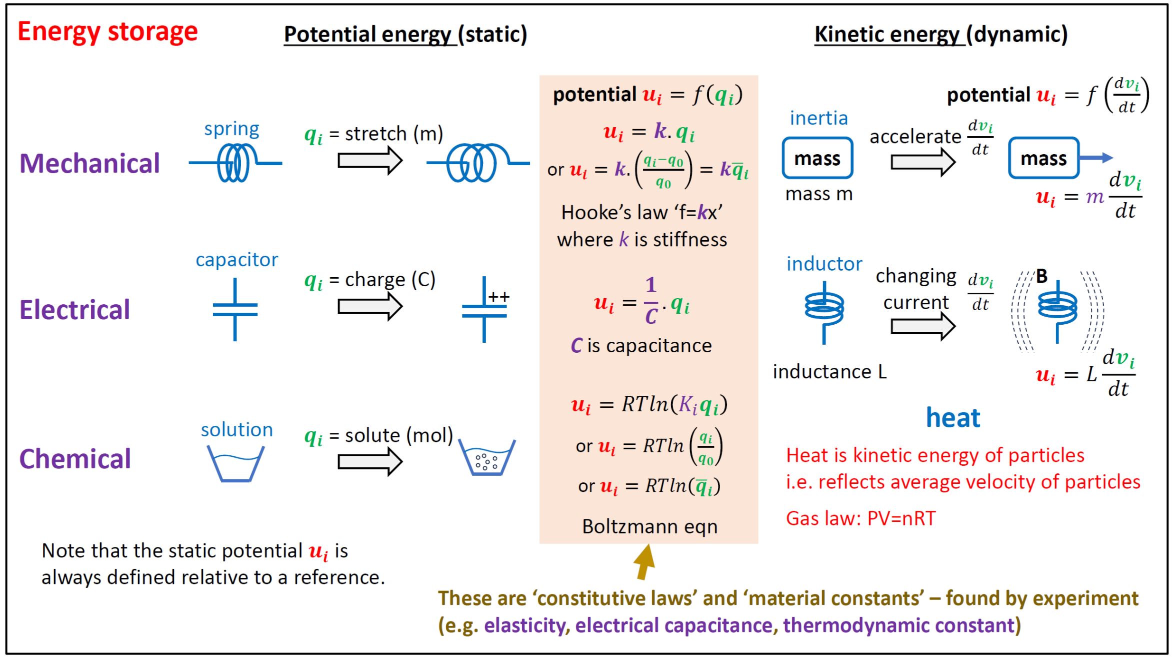
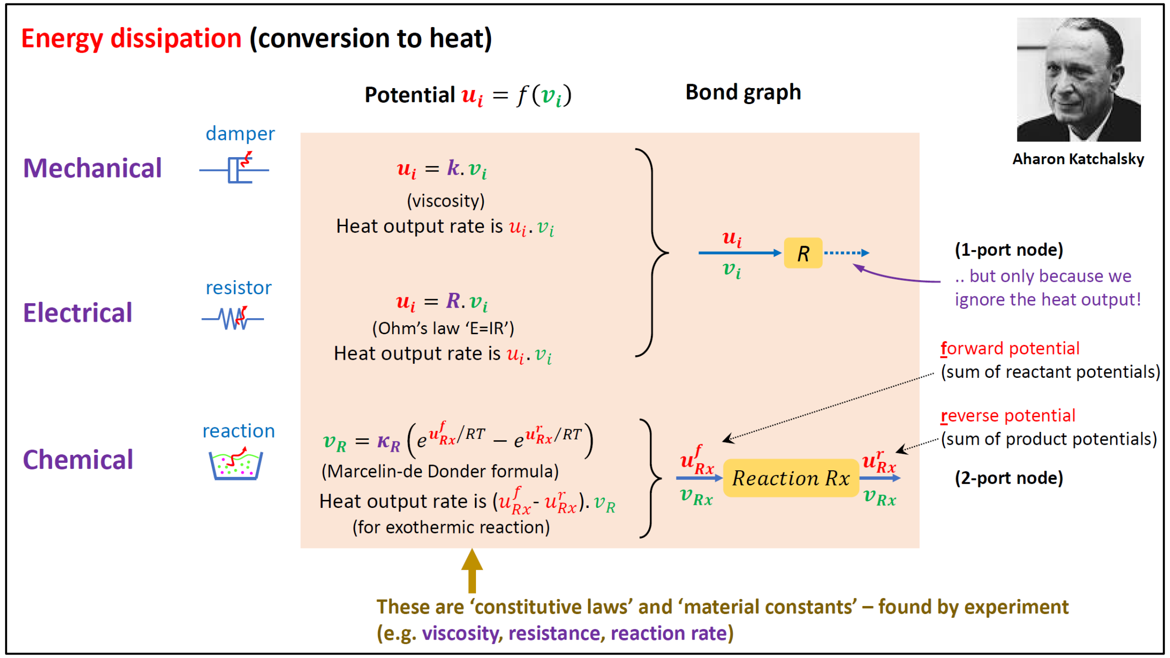
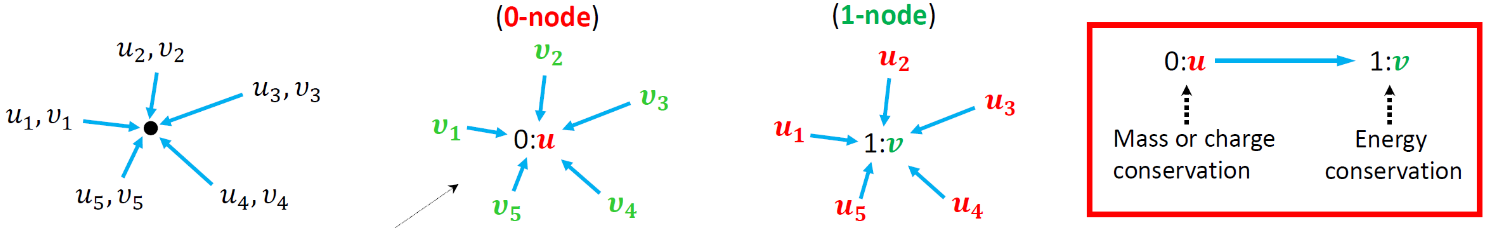
Create the environment in conda

- conda create --file pyenv_0D_DL.yaml

Bond graphs (a reminder from yesterday)





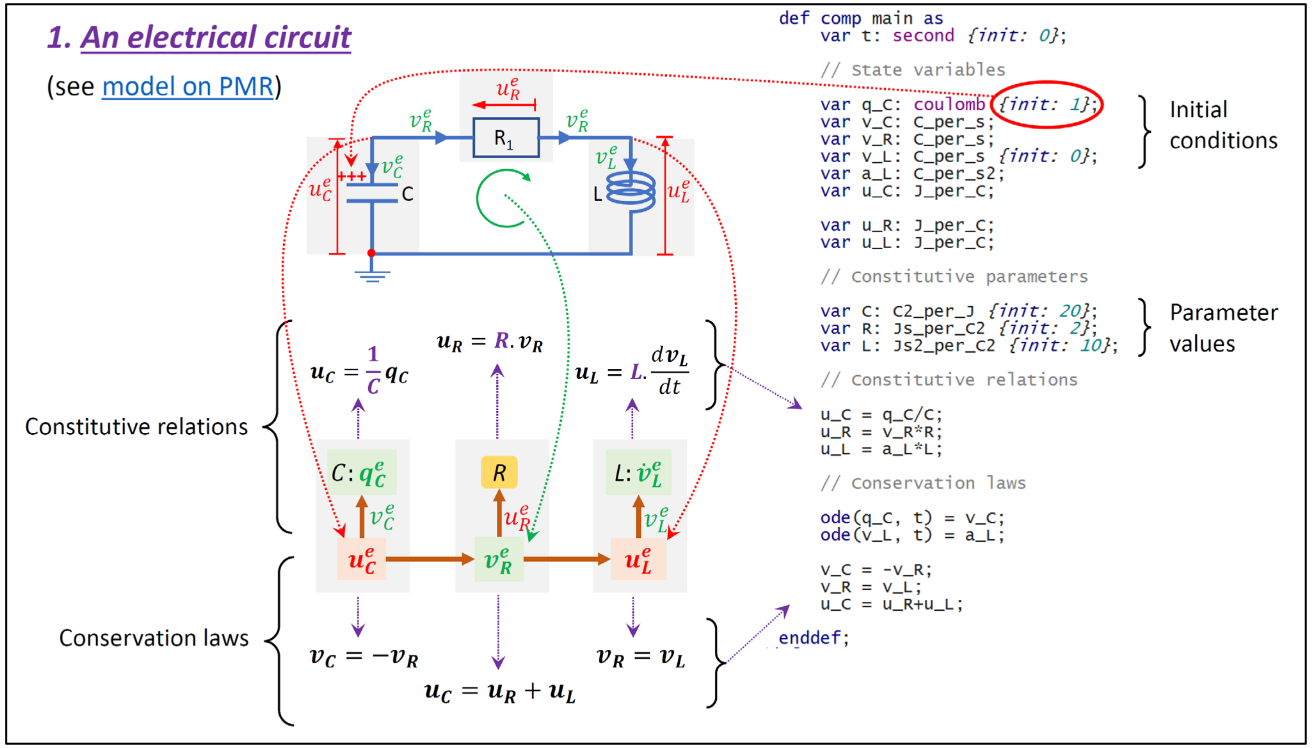


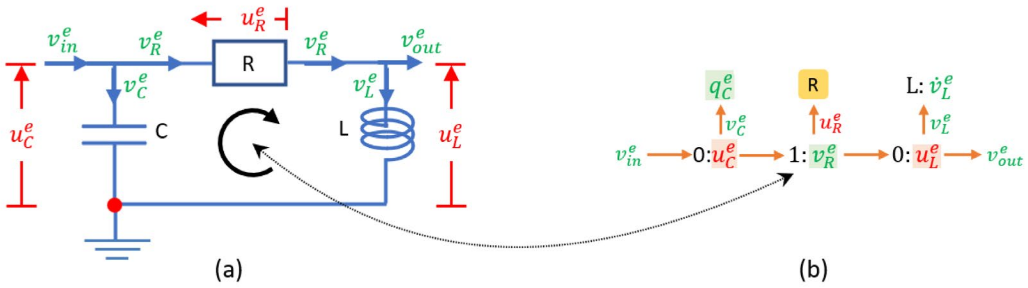
Note: To sum to zero, some of these flows must be negative!

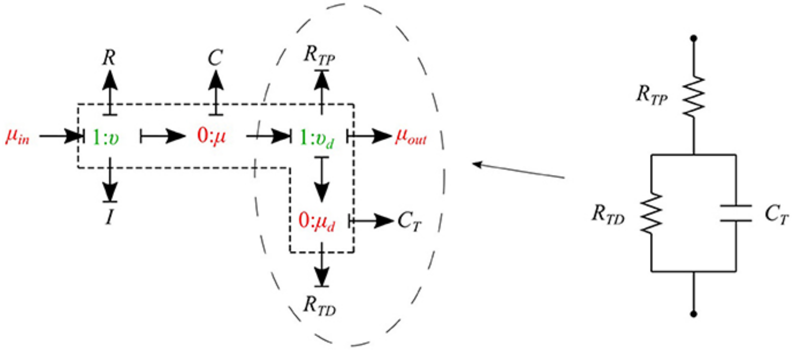
$\Sigma v_i = \frac{d}{dt}(\Sigma q_i) = 0$, so Σq_i is constant.

Taken from PJ Hunter's slides at TU Delft 18.03.2025

Bond graphs / Windkessel models







Safaei, Blanco, Müller, Hellevik, Hunter, *Frontiers in Physiology* 2018

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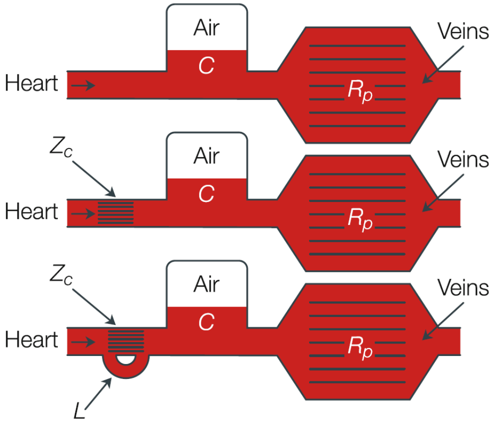
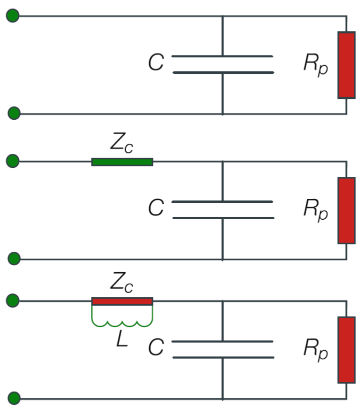
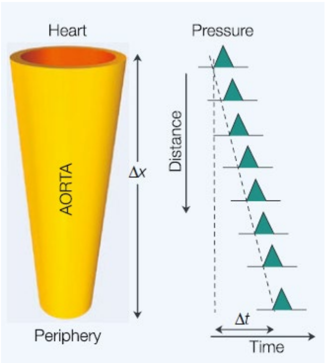
Schematics from PJ Hunter's slides at TU Delft 18.03.2025

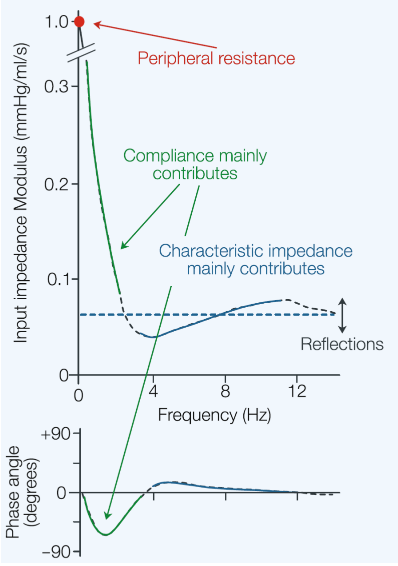
Basic notions





- Windkessel models
- 2-element: Total peripheral resistance (*R*), total arterial compliance (C)
- 3-element: R, C and aortic characteristic impedance (Zc)
- Pulse Pressure (PP) Method to estimate compliance
- based on the PP predicted by a 2-element Windkessel model of the arterial system
- doesn’t require entire waveform (unlike the decay time or the area method)
- Input & Characteristic Impedance
- Wave speed (Pulse Wave Velocity, PWV)





Schematics from the book Snapshots of Hemodynamics

Deep learning – an example

- From photoplethysmogram signals to age (regression or classification to young/elderly)
- Simple workflow – we are skipping hyperparameter tuning!